

How Does Gamification Affect Virtual Reality Nature Experience?

YUXUAN HUANG, University of Minnesota, USA

QIAO JIN, Carnegie Mellon University, USA

SARA LOTEPLIO, Colorado State University, USA

FRANCISCO R. ORTEGA, Colorado State University, USA

VICTORIA INTERRANTE, University of Minnesota, USA

Studies have shown that simulated nature in virtual reality (VR) can achieve restorative effects similar to real nature. To avert boredom and enhance existing virtual nature experience, researchers are exploring the potential of incorporating interactive and gamified elements to promote engagement and effectiveness. While relaxation games have restorative potential, prior studies did not consider how a gamified objective can affect mental load and whether it would interfere with “soft fascination”, which is the key mechanism of nature restoration according to the attention restoration theory. This study investigated the impact of including a gamified objective on VR nature experiences with respect to both psychological and cognitive restorative outcomes and user preference. We conducted a two-session within-subjects experiment comparing two conditions with similar nature environment and identical interaction features. In the hard fascination condition (HF), participants were instructed to find and take pictures of target objects; in the soft fascination condition (SF) they were instructed to freely explore and take pictures of whatever they liked. The quantitative results showed that incorporating goal-directed gamified tasks in VR nature experience could enhance cognitive performance but reduce relaxation benefits. The qualitative findings from semi-structured interviews triangulated the quantitative results and revealed additional benefits of gamification, such as directing user attention and promoting long-term engagement. The findings suggest that gamifying a VR nature experience introduces a tradeoff between relaxation and other benefits, such as engagement and cognitive performance. As such, gamification may be most appropriate in contexts where these alternative benefits are prioritized over relaxation alone.

CCS Concepts: • **Human-centered computing** → **Virtual reality**; **Empirical studies in HCI**.

Additional Key Words and Phrases: Virtual reality, virtual environment, nature immersion, restoration, relaxation, cognition

1 Introduction

Researchers have long documented the benefits of both extended immersion and brief exposure to natural outdoor environments, such as forests and parks [75, 77]. Nature immersion can reduce stress and negative emotions, enhance positive emotions and both mental and physical well-being [32, 59, 76, 78]. In situations where visiting nature environments in real life is not feasible, such as elderly in care homes, inpatient in hospitals, or for other individuals without easy access to nature, simulated nature contact via pictures, paintings, videos, etc., can also provide psychological and physiological benefits [11, 22, 42].

More recently, virtual reality (VR) technology emerges as one of the most popular media for simulated nature content thanks to the wider adoption of consumer VR devices and their unmatched immersion [65, 79, 83].

Authors' Contact Information: Yuxuan Huang, huan2076@umn.edu, University of Minnesota, Minneapolis, Minnesota, USA; Qiao Jin, Carnegie Mellon University, Pittsburgh, Philadelphia, USA, georgiej@andrew.cmu.edu; Sara LoTemplio, Colorado State University, Fort Collins, Colorado, USA; Francisco R. Ortega, Colorado State University, Fort Collins, Colorado, USA; Victoria Interrante, University of Minnesota, Minneapolis, Minnesota, USA, interrante@umn.edu.

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Fig. 1. The virtual city park environment. In the hard fascination condition (HF), participants completed a task of photographing target objects listed on info boards scattered around the environment; in the soft fascination condition (SF), they freely explored and photographed objects without a specific task.

However, there are still many open questions related to how to develop optimal VR-based nature experiences, including the choice among different types of VR content (e.g., 360/3D videos or computer generated 3D virtual environments [9, 85]) and the types of affordances for interacting with the simulated environment [13, 49].

Stephen and Rachel Kaplan's Attention Restoration Theory (ART) posits that nature immersion promotes restorative outcomes by offering opportunities for the recovery of depleted attentional resources. [39]. One of the most important mechanisms supporting this recovery is theorized to be the "soft fascination" that is effortlessly evoked by subtle dynamic phenomena like gently rippling water, swaying leaves, drifting clouds, etc., as well as by static qualities that invite prolonged undirected visual exploration, like richly detailed textures and organically varying colors. Although VR developers do their best to integrate these types of features into the virtual experience, the richness of the resulting experience is often lacking compared to the real-world counterpart, either due to the constrained viewpoint and the finite video length in pre-recorded 360/3D videos, or the limited model variability, texture resolution, and natural phenomena simulated in 3D virtual environments. Therefore, studies have shown that the benefits of simulated nature are usually to a lesser extent compared to their real-world counterpart [12, 56], and the long-term effectiveness or even willingness to re-engage with the same simulated nature content is also unproven [47, 63]. As a result, VR developers are prone to want to introduce gamified tasks into the nature immersion experience to avert boredom and promote an increased desire for prolonged or repeated engagement [49, 71]. While an immersive VR game may successfully evoke a strong sense of escape from one's daily routine and promote positive emotions, the goal-oriented nature of games at the same time introduces an element of "hard" fascination which is expected to interfere with the recovery of attentional resources that "soft" fascination promotes.

Inspired by this, our study aims to answer the research question: *To what extent does adding gamification enhance or impede the restorative effectiveness of a virtual reality nature immersion experience?* Through a mixed method approach, the goal of our study is to help the VR community understand the pros and cons of adding gamification to VR nature immersion experiences. Specifically, we focus on the impact of the goal-oriented aspects of gamification, rather than on the specifics of any individual game design. We hope that our findings will not only help to inform the design of more effective virtual nature relaxation applications but also to expand the understanding of how the ART applies in the context of virtual engagement in simulated nature environments.

2 Related Work

2.1 Attention Restoration Theory and Soft vs Hard Fascination

Among the many hypotheses explaining why nature environments are restorative, the attention restoration theory (ART) by Kaplan et al. is one of the most widely accepted and discussed [39]. According to the ART, “directed attention”, or focus, is the kind of attention that requires voluntary effort. Directed attention is crucial for effectiveness of the thoughts and actions of human, and is a resource that can be depleted by the cognitive demands of everyday life. Stress arises when a person’s available attentional resources are insufficient for the tasks they need to perform. Involuntary attention, or fascination, however, does not require voluntary effort. Fascination can be either hard or soft, where hard fascination is associated with low mental bandwidth (MB), e.g., playing video games, whereas soft fascination is associated with high mental bandwidth, e.g., taking a stroll in nature. The latter allows people’s mind to wander and their attentional resources to replenish while the fascination takes place [6]. The attention restorative theory proposes that fascination, along with being away, i.e., providing the sense of “getting away” from everyday life, extent, i.e., being coherent in a way that is easy to make sense of, and compatibility, i.e., meeting the need and preference of the individuals, constitute the four components of a restorative environment. Nature is restorative because in most scenarios it provides all four [40].

Many studies have attempted to examine the ART, particularly considering the restorative effects of nature experience on attention resource measured by various cognitive tests performance [57, 69]. It has been shown in many prior studies that nature exposure can have a significant positive effect on working memory performance, i.e., short term memorization capacity [4], and attentional control, i.e., the ability to control what you are focusing on [20]. In many of these studies, working memory was assessed by Forward or Backward Digit Span Tests [28, 58, 73], while attention control was evaluated through tasks like Stroop’s Colored Word Test [26, 34, 38], Necker Cube Pattern Control Score (NCPC) [16, 30], and Attention Network Test (ANT) [7]. Our work is also grounded in ART, and we investigated both the psychological aspects (e.g., mood, stress) and the cognitive aspects (e.g., working memory, attention control) of nature immersion in VR and how it might be affected by gamification.

2.2 Simulated Nature Immersion

There has been a long history of using virtual representations of nature environments as surrogates of their real-world counterpart for restorative purposes [79]. Despite not being explicitly considered as a relaxation activity, viewing virtual representations of nature environments, such as 2D picture slides [7, 8, 29] and nature-themed videos [46, 80] were very commonly used in studies examining the benefits of nature. Research has also shown that nature-themed artwork and imagery are highly preferred in hospitals settings and can improve patients’ well-being [14, 44]. These studies provided solid evidence that exposure to virtual nature content has both psychological and physiological benefits in similar ways as real nature environments.

More recently, VR-based nature experience has attracted increased interest. Compared to traditional 2D nature content, VR allows people to be fully immersed in the virtual environment, which offers an experience much closer to real-world nature exposure [47]. Research has also demonstrated the restorative effects of VR nature exposure in terms of psychological measures, e.g., Positive and Negative Affect Schedule (PANAS) [82], State-Trait Anxiety Inventory (STAI) [68], etc., and physiological measures, e.g., heart rate variability, electrodermal activity, etc [1, 2, 15, 52].

2.3 Interactive and Gamified VR Nature Immersion

While it is easy to imagine gently strolling for a fairly long time through a real world forest without getting bored, supporting sustained engagement with a VR-based nature immersion experience can be more challenging. Although 360 videos can capture rich visual detail [10, 13], they lack affordances for interactivity, including things as simple as changing the view when one moves one’s head. Therefore, 360 videos are prone to boredom

over repeated use because the experience would be completely identical each time. Besides, the inherent lack of 6DOF head tracking in 360 videos also increases the likelihood of inducing cybersickness, and that risk is further exacerbated if the user moves their head.

To counter these limitations, some researchers have explored the development of 3D computer-generated virtual natural environments, which support active locomotion [15, 52, 61, 72]. However, computer generated environments are intrinsically less detailed compared to real-world recordings, and even though users can actively explore in the 3D environment, they may still feel bored quickly if the environment remains unchanged.

Some work has also incorporated interactive elements into 3D VR nature immersion experiences to further increase stimulation and prevent boredom. For example, Bakk et al. [5] designed “Zenctuary VR” which features a simulated nature environment supporting various interactions with the nature elements, such as watering and gathering plants, organizing pots and stones, etc.

Yeo et al. [85] found that participants reported less boredom and better mood after engaging with an interactive 3D virtual underwater environment than after passively watching 2D or 360 videos of similar content. Martínez and Šimunić [51] performed a similar comparison between a VR experience allowing interaction with nature elements and 360 video viewing experience, and found the interactive experience led to more pronounced reduction of stress and tenseness.

Some preliminary attempts have been made to incorporate gamification elements into nature immersion experiences to make them more engaging, as prior research has shown that gamification can also be an effective means of facilitating relaxation and improving well-being in both traditional media [21, 37, 45, 54] and VR [17, 23, 43, 48]. For instance, Szczepańska-Gieracha et al. [71] developed a “Virtual Therapeutic Garden” intervention for patients with depressive symptoms, in which they would color a mandala in a virtual garden in VR. They found improvements in depressive symptoms relative to a control group who received fitness training and health-promoting education outside of VR. Liszio et al. [49] posited that adding gamification to a VR nature immersion experience could improve mood and enhance presence. They compared physiological stress, mood, anxiety and presence between a control condition and two modes of a VR experience at a virtual beach: an interactive mode with gamified tasks involving cracking coconuts, feeding turtles, and watering plants; and a non-interactive mode without the tasks. They found that relaxation and increase in positive affect were significantly more prominent in the gamified mode than the non-interactive mode and the control condition. Graf et al. [27] intentionally included cognitive tasks requiring memory and concentration in the form of mini-games in a VR nature walk experience for elderly people to enhance the stimulating effect. Although participants gave positive feedback about their experience during the interview, the pre-post changes in positive and negative affect were insignificant.

While interactions can enhance presence, and playing games can promote relaxation [36, 62, 81], it remains unclear whether the hard fascination elicited by goal-directed, gamified tasks may interfere with the soft fascination typically evoked by relaxation in natural environments [35]. This raises the concern that introducing gameplay could potentially diminish or even displace the restorative benefits associated with virtual nature exposure. Our study aims to assess the extent to which introducing gamification into a VR nature experience impacts various measures of psychological and cognitive restoration, as well as users’ preferences for future engagement. Given the broad scope of gamification and its many facets, this preliminary study focuses specifically on its goal-oriented aspect, since the inclusion of an objective is a key feature distinguishing gamified from ungamified VR nature experiences.

3 Methods

3.1 Study Design

We conducted a controlled user study to assess the impact of introducing gamification to VR nature restorative experiences. Our hypothesis was that adding gamification would deepen users’ sense of engagement with the

VR experience, and potentially heighten their feelings of presence in the virtual nature environment, possibly resulting in improved mood and a stronger sense of having “taken a break”, but at a potential cost of reduced attentional and stress recovery, due to engaging hard vs soft fascination.

Our experiment used a within-subjects design with two conditions: hard fascination condition (HF) and soft fascination condition (SF). The HF was characterized by the introduction of a gamified objective. Participants were told to look for 8 target objects in a city park environment and take pictures of them when found. In the SF, participants were instructed to leisurely explore the virtual environment, soaking in the atmosphere and taking photos of content they found interesting. Video illustrations and the specific instructions for participants in each condition are provided in supplementary materials. While there are infinite possibilities in how a VR nature experience could be gamified, we chose to implement a “treasure hunt” because we wanted to minimize potential confounding factors. Our current experimental design ensures that participants in both conditions saw the same things and interacted with the environment in the same way; the only difference was the introduction of a goal-oriented task. This allows us to focus more specifically on the essential theoretical question at hand, namely the impact of substituting hard for soft fascination. Since the scope of our study is VR nature for restoration, the hard and soft fascination were defined in a relative sense, and both conditions were designed to facilitate relaxation.

The study was divided into two sessions, one for each condition with at least 48 hours in between. Each session took less than an hour to complete. To mitigate potential boredom from repeated exposure to identical scenery in each session, we immersed participants in different, non-overlapping sections of the same large city park environment across their two sessions. The presentation order of both the conditions and the starting locations were counterbalanced across participants.

3.2 Participants

We recruited 32 participants (15 female, 17 male) aged between 18 and 38 ($M = 24.5$, $SD = 4.01$) via flyers, email lists, and social media. Demographic information, which was collected after the experiment, showed that five participants (16%) had never used VR before and 17 (53%) had only used it once or twice in total, while 5 (16%) reported using it occasionally, and 5 (16%) weekly or more. With regard to their prior experience with real world nature, 13 (40%) of participants reported regularly spending dedicated time enjoying the outdoors (multiple times/week), 8 (25%) often (20-50 days/yr), and 11 (34%) occasionally (about once/month). None reported doing so only rarely (once or twice per year). No one had specifically used VR for restoration or relaxation before. The participants received a \$25 gift card after completing both sessions of the study. The study was approved by the University’s Institutional Review Board (IRB).

3.3 Materials

Environment. The virtual environment used in this study is a modified version of the demo map in the *City Park Environment Collection LITE* from the Unreal Engine Marketplace (Figure 1). We chose this environment because it is vast in extent and close to photorealistic in detail and appearance. It features a wide variety of elements like trees, flowers and water features, that are characteristic of restorative natural environments [75, 77], and it represents a common site for outdoor relaxation. Within this virtual environment, we selected two nature-dominant regions with comparable features and spatial extent to serve as Environment A and Environment B. We placed fences on paths to delimit the boundary of the area the participants were supposed to stay in. In both environments, we introduced additional nature-themed elements, such as flowers and butterflies, that could serve as search targets in the HF condition and add visual interest in the SF condition. We avoided using non-natural elements as search targets in order to maintain a dominant focus on nature in both conditions. In the HF condition only, info boards displaying pictures of the 8 target objects could be found around the environment. Figure 2a) shows a close-up



Fig. 2. (a) The info board showing the 8 targets in the HF Condition. (b) The frame and reticle that indicates the camera is activated.

view of an info board, and figure 1 shows a side-by-side comparison of the same region in the city park in HF vs SF conditions, where the only difference is the presence/absence of the info board.

Interaction. During the VR experience, the participants wore the HMD and held a controller in their dominant hand while seated on a swivel chair. They could change their viewing direction by turning their head or the chair, and move forward in the direction they were looking at a speed of about 1.5 m/s by holding down the grab button of the controller. In both conditions, the controller doubled as a virtual camera that participants could use to take pictures to ensure that equivalent interaction capabilities were available in each condition. Pressing and holding the trigger would bring up a white frame and reticle (see figure 2b), and releasing the trigger caused a shutter sound to be played. In the HF condition, the frame and reticle changed color when the trigger was released, to indicate the validity of the selection: green if the virtual image contained a newly-found target object, yellow if the pictured target had already been captured, and red if no target was present. In the SF condition, the color of the frame and reticle remained white regardless of the image contents. Please refer to the supplementary video for a direct comparison of the color changes between the two conditions. In neither condition were participants able to look back over the photos they had taken.

Apparatus. The VR application was developed using Unreal Engine 5.3. The software ran on a workstation with an RTX 3080Ti graphics card, Intel i7-11700K 3.60GHz CPU, and 64G RAM. A Meta Quest 2 headset tethered to the workstation was used to deliver the VR experience to the participants.

3.4 Research hypotheses and measures

Informed by the ART and its recent elaboration by Basu et al. [6], our study seeks to experimentally assess the extent to which adding task-based goals to a VR nature immersion experience may enhance vs impede restorative outcomes and promote a stronger desire for sustained engagement with virtual nature interventions.

We considered multiple dimensions of restoration, including mood, stress, working memory capacity, and attentional control, as well as general perceived restorativeness. We also asked participants about the likelihood they would want to engage in each of our nature immersion experiences again, and, as a control measure, checked for cybersickness.

Our key hypotheses were that:

- Restorative outcomes would be higher, but only slightly so, in the SF vs HF condition.
- Desire for future engagement with the VR nature immersion app would be significantly higher in the HF condition

Below is a detailed description of each of the measures we used and how we intended for them to inform our investigations:

Stress. We directly quantified stress using the 6-item short form of the State-Trait-Anxiety Inventory (STAI-6) [50]. We applied a linear transform to map the STAI-6 scores to the range of STAI full scale, with 20 corresponding to no stress and 80 corresponding to the maximum level of stress.

Mood. We included mood in our outcome measures because some theories on the restorative benefits of nature immersion, like the SRT [78], assign a central role to enhanced mood as a driver of decreased stress and improved physiological state. To assess mood, we used the 10-item Positive and Negative Affect Schedule short form (PANAS-S) [74], in which 5 items measure positive affect (PANAS-P) and 5 measure negative affect (PANAS-N). The PANAS is one of the most commonly-used instruments in nature immersion research [53, 67]. Both PANAS-P and PANAS-N have a range of 5-25, where a larger value corresponds to stronger emotion.

Working Memory Capacity. According to the ART, the primary mechanism through which nature immersion produces restorative outcomes is via the replenishment of depleted attentional resources. However, attention is a multi-faceted concept, encompassing working memory, processing speed, and inhibitory control, and many different instruments have been used to assess different facets of attentional recovery in prior work [69]. In this study, we opted to assess working memory capacity using an auditory backward digit span (BDS) test [18], in which participants heard a series of digits being spoken by a computer app and they had to verbally report the same set of digits in the reverse order. The sequence length started at 4 and increased by 1 each time the participant answered correctly. After two incorrect trials in a row, the sequence length was decreased. We used the scoring method described in [84], in which a total of 10 trials are presented, and the average sequence length over all correct trials is computed.

Attentional Control. Prior work [66] has suggested that selective attention and cognitive flexibility are, along with working memory, among the cognitive abilities most improved after time spent in nature. Therefore, we decided to also include attentional control as one of our outcome measures. The Attention Network Test (ANT) [24] is a gold-standard instrument for assessing attentional control, but we feared that administering the ANT could be prohibitively time-intensive so, as some other researchers have done [3], we opted instead to use a short (40 trial) version of the Stroop Color Word Test [60], in which attentional control is reflected in the magnitude of the “Stroop effect” [70], i.e., the difference between mean reaction time in the congruent trials and incongruent trials.

Perceived Restorativeness. To assess participants’ overall subjective self-reported sense of restorativeness, we used a modified version [6] of the Perceived Restorativeness for Activity Scale (PRAS) [55]. We used the PRAS rather than the typical PRS [31] because our experimental manipulation did not involve the environments in which participants were immersed but rather the activities they engaged in during their immersion. The PRAS has a range of 5 to 45, where larger value corresponds to higher perceived restorativeness.

Desire for Future Engagement. As one of our main goals in adding gamification was to enhance participant's sense of engagement while immersed in VR nature and increase their desire for future engagement, we needed to check this explicitly. We did this by asking users to indicate, on an 11-point Likert scale, their level of agreement or disagreement with the single question: "If I had easy access to the application I experienced in session 1, I would be likely to use it again in the future", and same question for session 2.

Personal Preference. We also assessed participants' personal preference by asking them which condition they preferred more.

Cybersickness. We used the Cybersickness in Virtual Reality Questionnaire (CSQ-VR) [41] to check for cybersickness. We chose the CSQ-VR over the SSQ because in recent tests it has demonstrated better internal consistency than the SSQ and superior psychometric validity [41]. The CSQ-VR has a range of 6 to 42, where 6 corresponds to no cybersickness and 42 corresponds to maximum cybersickness.

Finally, we also conducted semi-structured exit interviews with our participants to get a deeper understanding of their experiences in each of our experimental conditions. In particular, we asked participants to tell us about the main differences in how they felt between the two VR sessions, potential improvements they'd suggest to the interactions and virtual environments, and any additional feedback they had. The interviews were audio recorded and transcribed for analysis.

3.5 Procedure

Figure 3 visually summarizes the main steps in our experimental procedure. In the first session, after reading the consent form and providing written consent, participants completed the STAI-6 and PANAS-S assessments to establish baseline measures of their stress levels and affect. Then they completed a brief training session to familiarize them with the BDS and Stroop tests, followed by baseline measures of their working memory capacity (BDS) and attention control performance (Stroop). After completing these baseline tests, participants performed 8 rounds of a dual n-back task [33], which served as a "cognitive stressor" to both deplete their attentional resources and to induce feelings of stress, increasing the potential for restoration to subsequently occur. During each round, participants were shown a sequence of 24 images, each presented in conjunction with an auditory signal. The images showed a red square in a random location on a 3x3 grid, and the auditory signal was a spoken random letter. Participants had to press a key labeled with an icon of a square when the location of the red square was the same as in the image two steps back, and a key labeled with an icon of a speaker when the letter they heard spoken was the same as they had heard two steps back ($n = 2$). The cognitive stressor lasted approximately 10 minutes in total. Afterward, participants repeated the BDS and Stroop tests to reassess their working memory capacity and attention control, and again completed the surveys of stress and affect. These pre-VR measurements served as a manipulation check for the stressor task, and provided a new baseline against which to assess participants' reported levels of stress and fatigue after the VR experience.

Then the participants put on the headset and the instructor read the instructions corresponding to the condition they were in. The text of these instructions can be found in the supplemental materials. The participants were instructed to stay on the path, and not cross the fences in the virtual environment. The participants spent 10 minutes in the VR nature environment, after which the screen faded to black. After the VR session, we again measured participants' stress levels (STAI-6) and affect (PANAS-S), as well as their working memory capacity (BDS) and attentional control (Stroop). Finally, we had participants complete surveys of perceived restorativeness (PRAS) and cybersickness (CSQ-VR). Note that the order in which we administered the psychological surveys and attentional tasks was different before and after the VR immersion. We had participants do the psychological surveys (STAI-6 and PANAS-S) immediately before and immediately after their VR experience, bookended by the attentional tasks, because the surveys take less time to complete, and we feared that the attentional tasks might

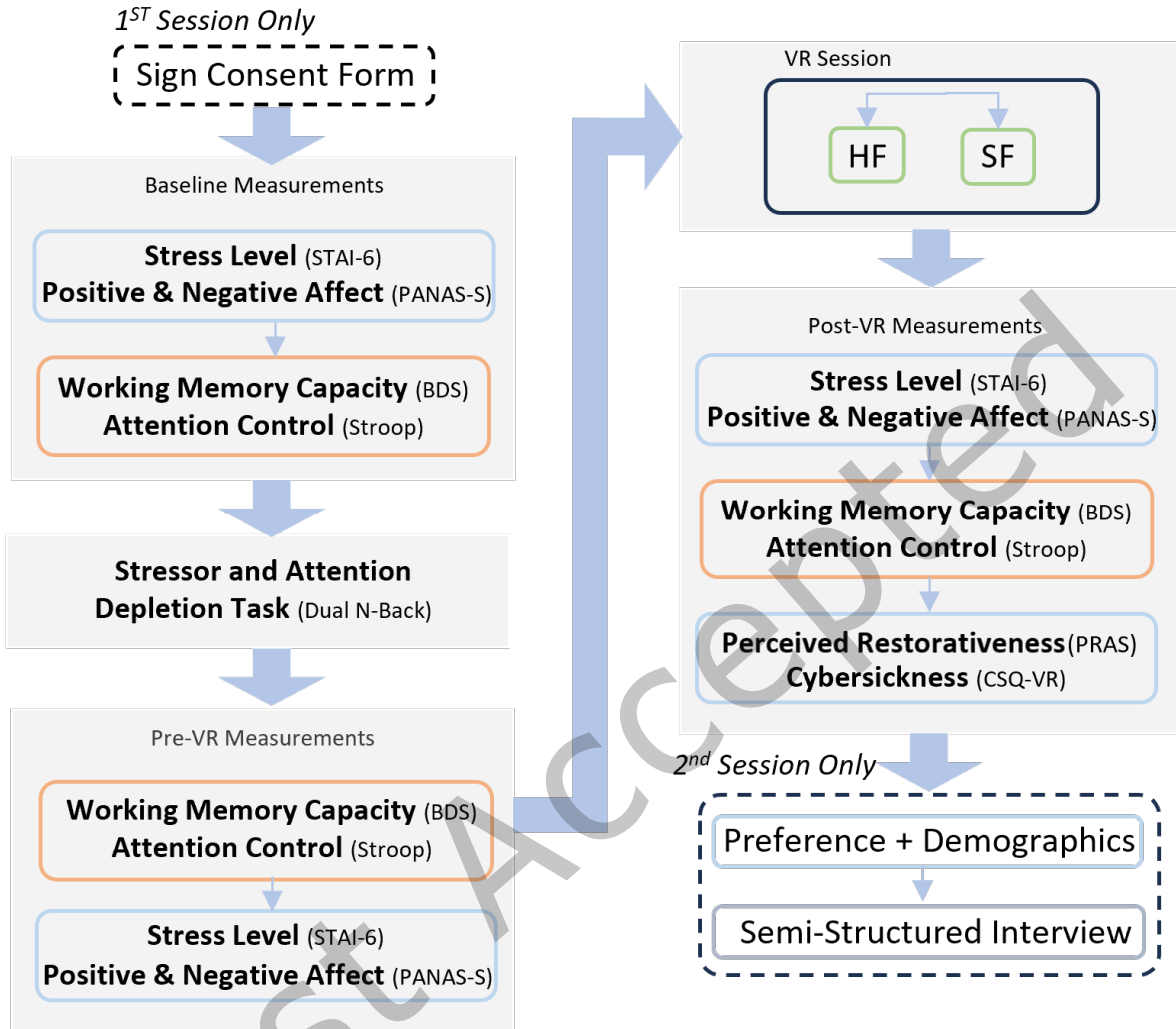


Fig. 3. The flowchart of the study procedure for the two session. The procedure was mostly identical between the two sessions, except that the informed consent procedure was only in the first session; the preference and demographics questionnaires, and the informal interview were only included in the second session.

themselves have a subtle negative impact on mood and stress, which we wanted to maximize prior to the VR experience but avoid potential interference from in our post-VR measures.

To minimize potential carry-over effects, we imposed a delay of at least 48 hours between each participant's two sessions. The second session was mostly identical to the first but without the consenting process or the training for the tasks. After the second session, we asked participants to complete an additional survey about their desire for future engagement with the VR app and to provide demographic data. We finished with a 5-10 minute structured exit interview.

3.6 Data Analysis

For the quantitative data, we began by using Cronbach's alpha to assess the internal consistency of each survey. If the score was below 0.7, we removed items with low correlations until Cronbach's alpha exceeded 0.7, indicating an acceptable level of internal consistency. To validate two assumptions of this study that 1) the cognitive stressor functioned effectively as a stressor and attention-depletion task, and 2) experiencing simulated nature in VR positively influenced stress levels, affect, working memory capacity, and attention control, we performed pairwise comparisons on the scores of the STAI-6, PANAS-S, BDS and Stroop Test between baseline and pre-VR (for assumption 1), and between pre-VR and post-VR (for assumption 2) for both HF and SF conditions, pooling data across both sessions. Specifically, we first used Shapiro-Wilk Test to test for normality. For normally distributed data we conducted paired T-Test. For non-normally distributed data we used Wilcoxon Signed-Rank Test. We averaged the scores across the two conditions for the comparisons between the baseline and pre-VR, since the assumption 1 was on the overall effects of the cognitive stressor instead of condition-specific effects.

To address our research question and uncover the differences between HF (with gamification) and SF (without gamification), we compared the changes in the scores of the above-mentioned measurements before and after VR, perceived restorativeness (PRAS), cybersickness (CSQ-VR), and user preference across HF and SF. We used mixed-effects linear regression because it effectively handles individual differences and allowed us to account for both conditions and session effects as well as the potential interaction effects between the conditions order (one limitation of counterbalancing [64]), providing a comprehensive analysis of the overall trends across conditions. Specifically, the model for each outcome measurement was specified as $Outcome \sim Condition \times Session$, where condition, session along with their interaction were included as fixed effects. Random intercepts were modeled for each participant, allowing us to account for repeated measurements within subjects. Additionally, a random slope for *condition* was specified within each participant to account for individual differences in how the outcome measurement was affected by HF vs SF. Moreover, to gain deeper insight into session-specific differences (i.e., first session was without carryover effects, second session was with prior experience for reference), we also performed pairwise comparisons separately for each session. We used a similar approach as noted before, except that we used independent T-test for normally distributed data, and Mann Whitney-U Test for non-normally distributed data due to the between-subject nature within each session.

For qualitative data, we applied a data-driven thematic analysis, drawing on grounded theory [19] to systematically code the data and identify key themes. The second author led the open coding process to maintain consistency, after which the first two authors collaboratively reviewed and grouped the codes, building an affinity map through constant comparison. In this process, each code was compared to existing codes and categories to find its best placement within the affinity map. Code placements were initially proposed by the first two authors, with any disagreements resolved through discussion, involving a third author when necessary. As higher-level themes emerged, the entire team engaged in discussion to evaluate the relevance, uniqueness, and contribution of each themes to the research goals. In this results section, we present the primary themes that surfaced from this process.

4 Results

Measures for stress, mood, working memory, and attention control were conducted 3 times in each session: at the beginning (baseline), after the attention depletion task and before the VR experience (Pre-VR), and after VR (Post-VR) (see figure 3). For consistency with the statistical tests used, we report mean (M) and standard deviation (SD) for all normally distributed data and visualize with bar charts. For all non-normal data, we report median (Mdn) and inter-quartile range (IQR) and visualize with box charts.

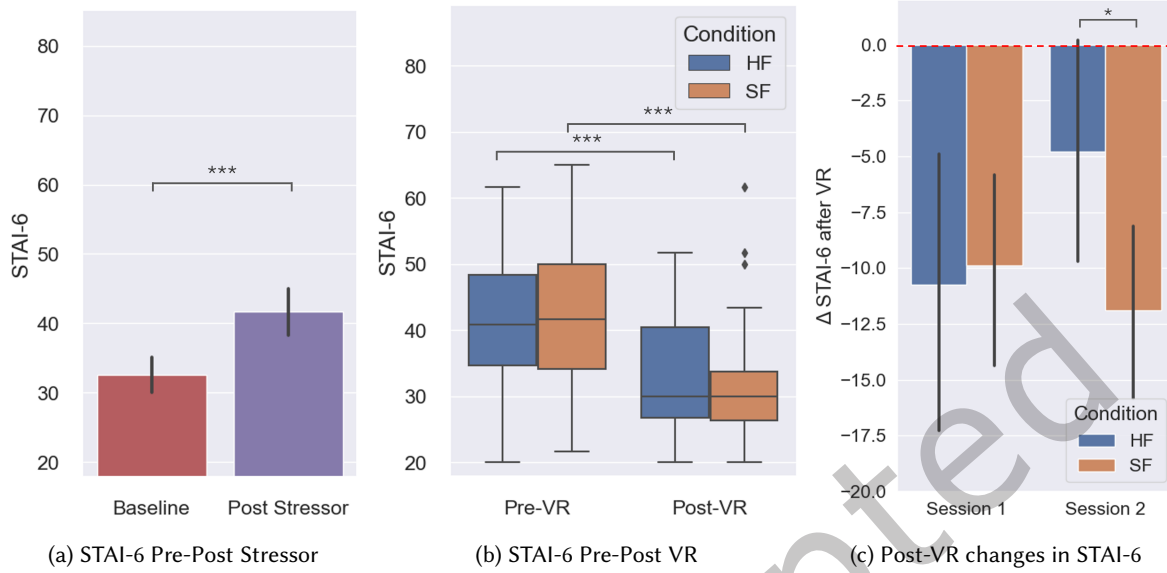


Fig. 4. (a) The STAI-6 scores at baseline and after the cognitive stressor. Overall the cognitive stressor increased stress. (b) The STAI-6 scores before and after VR exposure for each condition. The VR exposure reduced stress in both conditions. (c) Changes in STAI-6 scores after VR exposure in each session. In the 2nd session, SF led to significant more reduction in STAI-6 than HF after VR.

4.1 Stress

For STAI-6, the Cronbach's alpha was 0.782, indicating acceptable internal consistency.

Figure 4a shows the STAI-6 before and after the cognitive stressor. Paired T-Test showed that STAI-6 was significantly higher after the cognitive stressor ($M = 41.7, SD = 9.48$) compared to the baseline ($M = 32.53, SD = 7.12$) ($t = -6.41, p < 0.001$). This indicates that the cognitive stressor successfully evoked stress.

Figure 4b shows the STAI-6 before and after VR exposure. Wilcoxon Signed-Rank Test showed that in both conditions, STAI-6 was significantly lower after the VR exposure (HF : $Mdn = 30.0, IQR = 13.7$; SF : $Mdn = 30.0, IQR = 7.4$) than before (HF : $Mdn = 40.8, IQR = 13.7$; SF : $Mdn = 41.7, IQR = 15.8$) (HF : $W = 54.0, p < 0.001$; SF : $W = 8.5, p < 0.001$). This indicates that the VR experience effectively reduced stress.

The mixed linear model found no significant fixed effects for session ($z = 1.41, p = 0.16$), condition ($z = 0.22, p = 0.82$), or their interaction ($z = -1.45, p = 0.15$) on the changes in STAI-6 after VR exposure.

Figure 4c shows the changes in STAI-6 before and after VR separately for each session. In session 2, SF was associated with a significantly larger decrease in STAI-6 ($M = -11.88, SD = 8.23$) than HF ($M = -4.79, SD = 10.14$) after VR exposure ($t = 2.10, p = 0.04$), indicating that in the second session the participants reported more significant stress reduction in SF than HF.

4.2 Mood

4.2.1 Positive Affect. For PANAS-P, the Cronbach's alpha was 0.778, which was acceptable. Paired T-Test showed no significant difference in PANAS-P before and after the cognitive stressor ($t = -1.60, p = 0.12$), indicating that the cognitive stressor did not significantly affect positive mood. Wilcoxon Signed-Rank test found that the "alert"

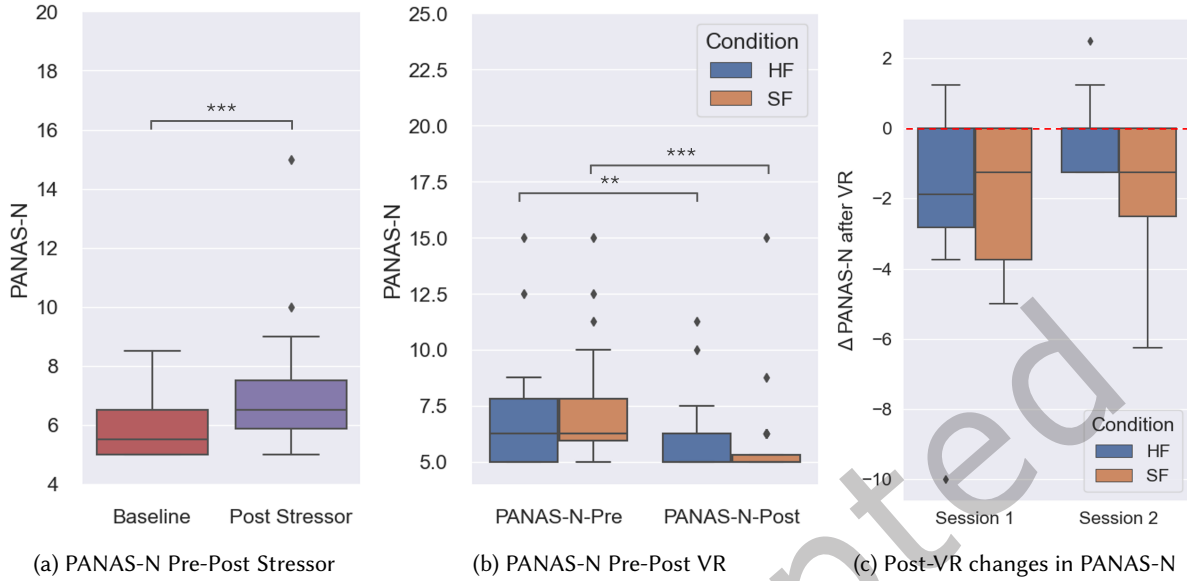


Fig. 5. (a) The PANAS-N scores at baseline and after the cognitive stressor. Overall the cognitive stressor increased negative affect. (b) The PANAS-N scores before and after VR exposure for each condition. The VR exposure reduced negative affect in both conditions. (c) Changes in PANAS-N before and after VR exposure in each session. Similar to STAI, in the 2nd session, SF led to more reduction in negative affect than HF after VR, but the difference was not significant.

subscale was significantly higher after the cognitive stressor than before ($W = 42.0, p = 0.005$). In both conditions, PANAS-P was significantly lower after VR exposure ($HF : M = 12.97, SD = 4.45; SF : M = 12.31, SD = 4.42$) than before ($HF : M = 14.31, SD = 4.75; SF : M = 13.44, SD = 4.96$) ($HF : t = 2.52, p = 0.02; SF : t = 2.08, p = 0.046$). The decrease in PANAS-P was driven by the subscale attributes “alert” ($HF : W = 127.5, p < 0.01; SF : W = 91.0, p < 0.01$) and “active” ($HF : W = 88.5, p = 0.02; SF : W = 132.0, p < 0.01$).

The mixed linear model for PANAS-P found no significant fixed effects for session ($z = 1.36, p = 0.17$), condition ($z = 0.64, p = 0.52$), or their interaction ($z = -0.49, p = 0.63$) on the changes in PANAS-P after VR exposure.

4.2.2 Negative Affect. For PANAS-N, the Cronbach’s alpha was 0.699. We dropped ‘hostile’ to increase it to 0.707, and rescaled the resulting score to the original scale.

Figure 5a shows the PANAS-N scores before and after the cognitive stressor. Wilcoxon Signed-rank Test showed that PANAS-N was significantly higher after the cognitive stressor ($Mdn = 6.5, IQR = 1.63$) compared to the baseline ($Mdn = 5.5, IQR = 1.50$) ($W = 28.0, p < 0.001$). This indicates that the cognitive stressor successfully induced negative mood.

Figure 5b shows the PANAS-N scores before and after VR exposure. In both conditions, PANAS-N was significantly lower after the VR exposure ($HF : Mdn = 5.0, IQR = 1.25; SF : Mdn = 5.0, IQR = 0.31$) than before ($HF : Mdn = 6.25, IQR = 2.81; SF : Mdn = 6.25, IQR = 1.88$) ($HF : W = 26.0, p < 0.01; SF : W = 0.0, p < 0.001$). This indicates that similar to stress, the VR experience effectively reduced negative mood.

The model for PANAS-N found significant fixed effects for session on the changes in PANAS-N after VR exposure ($estimate = 1.88, SD = 0.70, z = 2.67, p < 0.01$), indicating less decrease in PANAS-N after VR in the

second session. However, no significant effects was found for condition ($z = 0.35, p = 0.73$), or the interaction between session and condition ($z = -1.34, p = 0.18$).

Figure 5c shows the changes in PANAS-N before and after VR separately for each session. PANAS-N exhibited a pattern similar to STAI-6. In the second session, SF was associated with a larger decrease in PANAS-N ($Mdn = -1.25, IQR = 2.5$) than HF ($Mdn = 0.0, IQR = 1.25$) after VR exposure ($U = 174.0, p = 0.07$), while the significance did not reach the 0.05 threshold.

4.3 Working Memory

Figure 6a shows the BDS scores before and after the cognitive stressor. Paired T-Test showed that the BDS score was significantly higher after the cognitive stressor ($M = 7.81, SD = 1.33$) than before ($M = 7.36, SD = 1.18$) ($t = -3.60, p = 0.001$).

Figure 6b shows the BDS scores before and after VR exposure. No significant difference was found in BDS score before and after the VR exposure in either HF ($t = -1.50, p = 0.14$) or SF ($t = -0.89, p = 0.38$).

The mixed linear model found significant fixed effects for session ($z = -2.02, p = 0.04$), condition ($z = -1.96, p = 0.049$) and their interaction ($z = 3.099, p < 0.01$) on the improvement in BDS score after VR immersion. The result suggested a significant decrease in BDS improvement from HF to SF ($estimate = -0.80, SD = 0.41$) and from session 1 to session 2 ($estimate = -0.87, SD = 0.43$). The interaction ($estimate = 1.26, SD = 0.41$) implied that the difference between HF and SF reduced in the second session.

Figure 6c shows the changes in BDS scores before and after VR separately for each session. Paired T-Test showed that in the first session there was a significant improvement in BDS score after VR immersion in HF ($t = -2.62, p = 0.02$), but not in SF ($t = 0.11, p = 0.92$). Independent T-Test also showed significant difference between HF and SF in the changes in BDS score before and after the VR session ($HF : M = 0.77, SD = 1.14; SF : M = -0.03, SD = 0.94; t = 2.09, p = 0.045$). However, the trend was not present in the second session. This result reinforced our finding in the mixed linear model, indicating that HF was more effective in improving working memory, particularly in the first session.

4.4 Attention Control

Figure 7a shows the Stroop effect before and after the cognitive stressor. Paired T-Test showed that the Stroop effect was significantly larger after the cognitive stressor ($M = 97.50, SD = 65.17$) than before ($M = 70.50, SD = 71.58$) ($t = -2.08, p = 0.046$), indicating a decline in attention control.

Figure 7b shows the Stroop effect before and after the VR exposure. No significant difference in Stroop effect was found before and after the VR exposure in either HF ($t = 0.57, p = 0.57$) or SF ($t = -0.42, p = 0.68$) conditions, indicating that neither VR experience demonstrated significant impact on attention control overall.

The mixed linear model found no significant fixed effects for session ($z = 0.90, p = 0.37$), condition ($z = 1.09, p = 0.28$), or their interaction ($z = -0.84, p = 0.40$) on the difference in Stroop effect before and after the VR session.

Figure 7c shows the changes in Stroop effect before and after VR separately for each session. We also did not see any significant difference in the reduction of the Stroop effect after VR exposure between conditions. Mann Whitney-U Test showed a similar trend as in the BDS score, where in the first session HF was associated with larger decrease in the Stroop effect ($Mdn = -51.0, IQR = 138.75$) after VR exposure compared to SF ($Mdn = 48, IQR = 160.25$). However, the significance did not reach the 0.05 threshold ($U = 83.5, p = 0.097$).

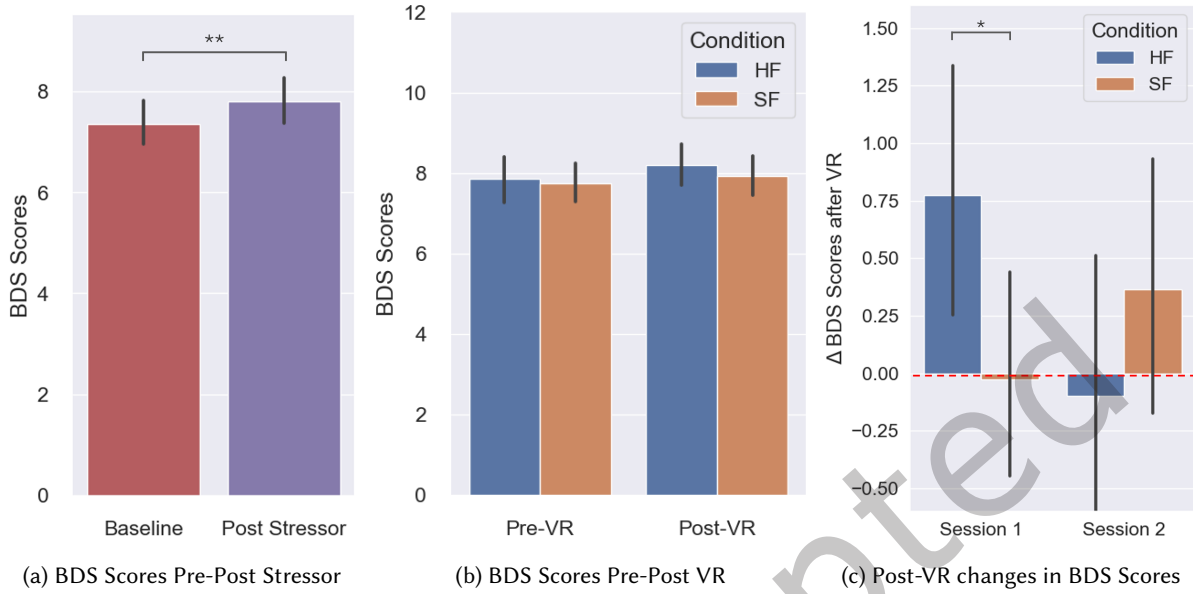


Fig. 6. (a) The BDS scores at baseline and after the cognitive stressor. The cognitive stressor led to improved working memory. (b) The BDS scores before and after VR exposure in each condition. No significant difference was found (c) Changes in BDS scores before and after VR exposure in each session. In the first session, HF led to significantly more improvement in BDS than SF after VR exposure.

4.5 Perceived Restorativeness

For PRAS, the Cronbach's alpha was 0.925, indicating high reliability. Wilcoxon Signed-rank Test showed no significant difference in PRAS between HF ($Mdn = 32.0, IQR = 6.5$) and SF ($Mdn = 30.0, IQR = 11.25$) ($W = 215.5, p = 0.72$).

The mixed linear model found no significant fixed effects for session ($z = 0.33, p = 0.74$), condition ($z = -0.38, p = 0.70$), or their interaction ($z = 0.10, p = 0.92$) in the PRAS responses. Independent T-test also showed no significant difference between them within either session 1 ($t = 0.40, p = 0.69$) or session 2 ($t = 0.19, p = 0.85$).

4.6 Preference and Desire for Future Engagement

Out of the 32 participants, 21 preferred HF, and 11 preferred SF, which showed an overall preference towards HF, but a binomial test showed that the observed frequency of 21/32 was not significantly different from 1/2 ($p = 0.11$). However, we noticed a strong difference in preference patterns between those experiencing the two conditions in different orders: of the 16 who experienced HF first: 7 (44%) preferred HF and 9 (56%) preferred SF; of the 16 who experienced SF first: 2 (12%) preferred SF and 14 (88%) preferred HF. This can be explained by a greater tendency to report preferring whatever was done more recently, in combination with a preference for HF over SF.

Wilcoxon Signed-rank Test also showed no significant difference between HF ($Mdn = 7.0, IQR = 6.0$) and SF ($Mdn = 8.0, IQR = 5.25$) ($W = 136.5, p = 0.31$) in participants' 11-point likert-type responses to how likely they will be using the application again in the future.

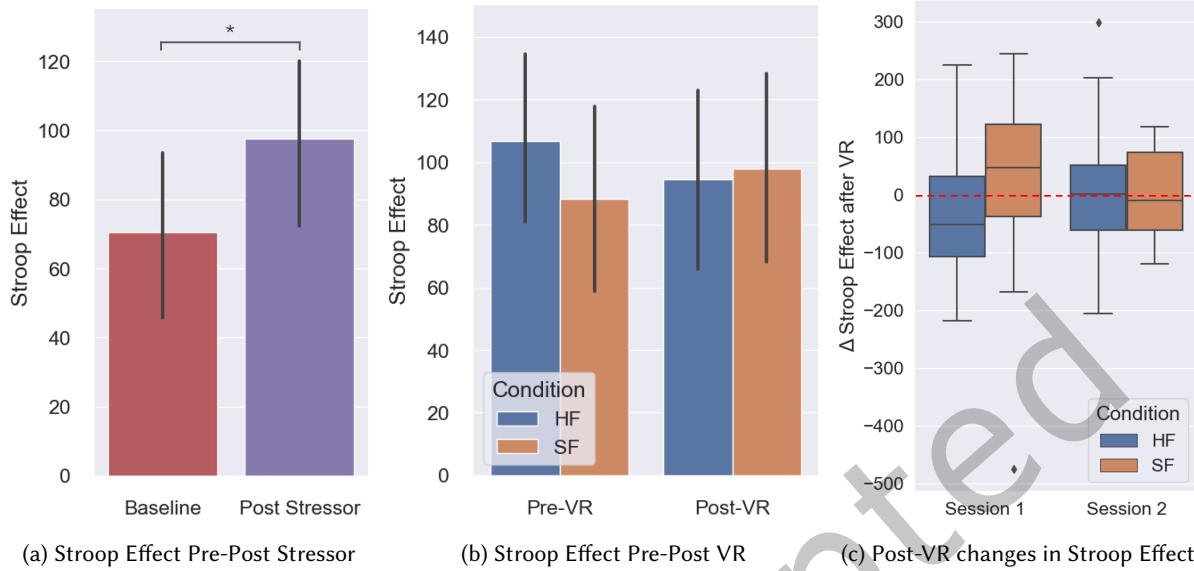


Fig. 7. (a) The Stroop effect at baseline and after the cognitive stressor. The cognitive stressor led to reduced attention control. (b) The Stroop effect before and after VR exposure in each condition. No significant difference was found. (c) Changes in Stroop effect before and after VR exposure in each session. Similar to BDS, in the second session, SF led to more reduction in stroop effect than HF, but the difference was not significant.

4.7 Cybersickness

For CSQ-VR, the Cronbach's alpha was 0.6431, which could not be improved by removing items. The cybersickness level was low and for both HF ($Mdn = 8.5$, $IQR = 5.25$) and SF ($Mdn = 10.0$, $IQR = 5.0$) conditions. The range of CSQ-VR is 6 to 42, with larger value indicating more severe cybersickness. Wilcoxon Signed-rank Test found no significant difference between HF and SF ($W = 137.5$, $p = 0.50$).

4.8 Qualitative Results

4.8.1 Theme 1: Gamification averts boredom but may interfere with relaxation. Gamification elements in VR nature increased user engagement by offering clear goals that arouse interest. In general, participants appreciated how goal-directed exploration (HF) kept them engaged during the restoration process and prevented boredom. Many participants noted that they felt bored after a few minutes of just walking around in SF, while some of them described the free exploration as "somewhat boring but relaxing and calming" (P32). The presence of the task in HF made participants more attentive compared with the "passive wandering" (P22) in SF.

While gamification increased engagement, it also risked "disrupting the relaxing nature atmosphere" (P6). Participants indicated that HF, though engaging, could sometimes made them "feel anxiety, though not to the point super stressful" (P6) and "prevent full enjoyment of the nature" (P3). P8 felt a sense of time pressure to complete the task, even though finding all targets was not required. The balance between engagement and relaxation was delicate—while some found the gamified condition more active and less boring, others felt it increased pressure and reduced the calming effect of the virtual nature setting. To address the conflict, P6 highlighted the importance of "an appropriate intensity" to prevent the tasks from becoming too demanding or dominating the experience.

4.8.2 Theme 2: Gamification reduces awareness of environmental details unrelated to the task. Most participants noted that multi-sensory details—like spatial audio, foliage animations, and realistic graphics—enhanced immersion and the restorative effects of the VR nature experience. Some participants reported that the gamified task in HF shifted their focus away from their awareness of the sensory stimuli unrelated to the task. For example, P15 was unaware of the spatial audio until the second session (SF), and P17 paid more attention to details such as the stem of the lotus under the water, etc. in SF.

On the other hand, overlooking unrelated details also made the unrealistic/undesirable elements less noticeable. For instance, P15 mentioned being “less aware of the slow movement speed” in HF, even though the speed was the same in both conditions. P5 reported being less aware of the LOD (Level of Detail) popping in HF, where the abrupt changes in visual quality were less noticeable, making the experience smoother and more immersive compared to SF. In addition, participants also noted that the gamified task led to more awareness of task-relevant aspects of the environment. Several participants noted that task-based objectives motivated them to engage with interesting areas or details they might have otherwise overlooked. For example, P2 remarked, “I was looking at it (the environments) more carefully and I saw different things. Like in the first session (SF), I did not see the butterfly.”

4.8.3 Theme 3: Gamification increases willingness to use VR for restoration and has the potential to boost long-term appeal. Most participants acknowledged that both conditions offered restoration benefits, but when asked about future use, many highlighted that the gamification in the VR nature experience increased their willingness to use it, as it provided a clear “purpose” (P27, P32). Several users noted that without gamification elements, they would be less motivated to use VR as a tool for relaxation. As P19 remarked, “If using VR is just to casually look around, I’d rather not use it and just take a rest for 10 minutes.” When gamified elements were incorporated, even with simple tasks like fishing (P1) or flower-picking (P20), participants expressed greater enthusiasm to use VR for restorative activities.

When asked about revisiting the same virtual environments for restoration, many participants felt it would lose long-term appeal if the environment remained unchanged, especially in SF. Both P7 and P31 noted that for HF, it is acceptable to revisit the same nature environment as long as there are new targets to find. However, for SF, visiting the same area multiple times would quickly become boring and the motivation to use it again would reduce significantly. Thus, gamification introduced greater potential for repeated/long-term use of VR nature environments by offering “interesting tasks that they want to do multiple times” (P15).

5 Discussion

In this section we discuss the effects of gamification on VR nature experience and provide recommendations on developing and gamifying VR nature experience. It is worth emphasizing again that implementing different games in a VR nature setting can lead to drastically different results. Our study is designed to focus on the influence of the goal-directed nature of gamification by keeping the visual and interactive features consistent across the gamified and non-gamified conditions. We consider the influence of this goal-directed component to be generalizable to other game implementations, but the influence of gamification as a whole is not generalizable, as it depends heavily on the specific design and implementation of the games.

5.1 Effects on Psychological Restoration

Stress reduction and emotional regulation is considered one of the main benefits of nature immersion. Both the HF and SF were shown to be effective in reducing the stress and negative emotions induced by the cognitive stressor from our quantitative results. This confirmed the positive effects of VR nature exposure on emotional regulation, which was consistent with findings from prior studies [25, 47].

While both conditions facilitated relaxation, both quantitative and qualitative results indicated more relaxation in SF than HF. The survey responses revealed a larger reduction in stress after VR in SF than in HF in the second session, and a similar trend for negative affect, although it did not reach statistical significance. The lack of significant differences in stress and negative affect reduction between conditions in the first session was likely due to the participants not having a reference for scoring their experience, as we anticipated in section 3.6. Qualitative responses further supported the quantitative findings, with several participants reporting that they felt more relaxed in the absence of the gamified task (Theme 1). Together, these findings support our concern that incorporating a gamified goal in VR nature experiences may interfere with the “soft fascination” that nature typically evokes. However, when the intensity of the game is kept low, e.g., as in this study, the interference appears to be minimal and does not entirely negate the psychological restorative benefits of VR nature exposure.

5.2 Effects on Cognitive Performance

Attention restoration is another important aspect of the benefits from nature according to ART. Both the mixed linear regression and the pairwise comparison in the first session showed that HF led to significantly more improvement in BDS performance than SF. A similar trend was also observed in the Stroop effect despite not reaching the significance threshold. This result was also echoed by several participants’ interview responses (e.g., P6 reported that they did not feel ready for the cognitive tests after VR in SF because they were in a very relaxed state, whereas in HF they were more active throughout, thus more ready for subsequent tests). These results implied that HF was more beneficial for cognitive performance than SF.

According to the ART, SF would be expected to be more beneficial for cognitive performance since it occupied less mental bandwidth. However, in our study since the participants’ attention resource was not depleted by the cognitive stressor before entering VR, the Yerkes-Dodson Law [86] might explain the results better. According to the law, the relation between arousal level and task performance follows a bell-shaped curve, where task performance increases with arousal level but only up to a point, after which the performance decreases as arousal level goes up. In our case, both HF and SF were designed to be relaxing experiences and were unlikely to cause over-arousal. Compared to SF, HF was expected to be associated with higher arousal level due to the gamified tasks, which explained why HF was associated with higher cognitive test performance.

The difference between the two conditions in cognitive improvement diminished in the second session, which might be due to the carryover effect mentioned in section 3.6. In the second session, participants who experienced HF had already experienced SF in their first visit, which was similar in many ways, so the arousal level might not be the same as the other group of participants when they experienced HF in their first session for the first time.

5.3 Additional Effects of Gamification

Besides the effects on psychological and cognitive restoration, the qualitative results revealed additional effects of incorporating a gamified objective in VR nature exposure. In Theme 1, participants noted the tradeoff between relaxation and averting boredom. Despite feeling less relaxed compared to a free exploration, the presence of a gamified task gave them a sense of purpose to stay in the virtual environment, which increased the willingness to use the application for many participants. This is related to Theme 3, which highlighted the potential of gamification to enhance long-term appeal by keeping users engaged over multiple sessions. Although repeatedly playing the same game over an extended period may eventually lead to boredom, this issue might be effectively mitigated by introducing gameplay variations (e.g., adding new targets to the treasure-hunting task), which is significantly more cost-effective than developing entirely new virtual environments.

Theme 2 offered additional insight that the inclusion of a task directed participants’ attention onto specific aspects of the environment related to the task. This leads to participants being less aware of the computer graphics

artifacts of the virtual environment, and allows designers to have more control over how users experience VR nature.

5.4 Recommendations on Designing VR Nature Applications

Based on the results of the study, the overall difference in psychological and cognitive benefits between the HF and SF was subtle, but the presence of a gamified objective did introduce a tradeoff between relaxation, arousal, and engagement. This provides important implications for the design of VR nature applications.

In cases where a full relaxation is the sole goal, the presence of a gamified objective in VR nature experiences can be counter-productive. However, this must be weighed against the motivational benefits that gamification can offer. Boredom or a perceived lack of purpose in the virtual environment may reduce users' willingness to engage with the application—particularly over the long term. Our findings suggest that the majority of the restorative benefits can still be preserved with gamification, provided the task remains low in arousal. Conversely, when the goal is brief relaxation in preparation for an upcoming task, incorporating gamified elements may help prevent under-engagement while still delivering meaningful psychological restoration. Therefore, gamification is best suited for scenarios where engagement, stimulation, or sustained use is prioritized over pure relaxation.

6 Limitations and future work

This paper focuses specifically on the effects of introducing a gamified objective, which represents a key distinction between gamified and ungamified VR nature experiences. However, this focus does not capture the full impact of gamification, which is influenced by a wide range of design factors. Future research should explore more nuanced gamification mechanisms, such as game types, reward structures, and social components, to develop a clearer understanding of how to optimize the design of VR nature experiences for different use cases.

Additionally, the study examined only a single type of natural environment, i.e., a city park, and primarily involved younger participants (18–38 years old). Therefore, the generalizability of the findings should be further validated across a broader range of natural settings, such as forests and mountains, as well as among more diverse populations, including older adults who may particularly benefit from virtual nature experiences.

While participants highlighted the potential long-term advantages of gamification in their interview responses, confirming these benefits requires a longitudinal study to examine how user retention and engagement develop over time in both gamified and ungamified VR nature experiences. Similarly, future studies can also consider how prior VR experience affects the restorative benefits obtained from gamified vs ungamified experience, since a novice user may be more “fascinated” by VR nature compared to an experienced VR user even without gamification, thus affecting restorative outcome.

Our cognitive stressor did not lead to reliable attention fatigue. Prior studies have demonstrated that nature exposure can lead to improved performance in BDS and Stroop even without introducing restorative potential with attention-draining tasks[69], so this should not affect the validity or meaningfulness of our results. However, our results regarding the cognitive benefits should be interpreted as cognitive performance improvement rather than restoration of depleted attention. For future work, it is worth conducting longer attention draining tasks from 30 minutes to an hour to introduce more significant attention fatigue in participants to see if the cognitive advantages of a gamified VR nature experience persist in that setting.

Lastly, the gamified task in this study was designed to be nature-themed, i.e., searching for nature elements, which may not be generalizable to gamified tasks unrelated to nature elements. Future studies can also examine whether the restorative benefits of nature persist if the gamified task draws people's attention to non-nature elements, or if it would be “a waste of a break” [35].

7 Conclusion

In this paper, we investigated how gamification affects VR nature exposure by focusing on its core distinction from ungamified experiences: the presence of a gamified objective. We conducted a within-subject study comparing two nearly identical VR conditions, differing only in the inclusion of a gamified search task. The results showed that the gamified condition elicited higher arousal and improved cognitive test performance, whereas the ungamified condition was more effective for relaxation and for alleviating stress and negative emotions. Qualitative feedback further revealed that gamification may offer additional indirect benefits, such as increased willingness for long-term or repeated use and reduced awareness of graphical imperfections in the virtual environment. The results highlight a tradeoff introduced by gamification in VR nature experiences: while it enhances engagement and cognitive stimulation, it may do so at the expense of restorative outcomes. Accordingly, gamified designs may be more appropriate in contexts where long-term or frequent use is expected, or when active participation is valued over sheer restorative effects.

This study serves as an initial exploration of gamification's impact on VR nature experiences by focusing exclusively on the presence of an objective, though its findings may not generalize across all forms of gamification, as different game mechanics and designs could yield varying effects. Future research should examine a broader range of gamification elements to develop more nuanced design guidelines.

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